

Bonus 3 - řešení

① Autonomní rovnice:

$$y' = \sqrt{y^2 + 3}$$

$$\frac{dx}{dt} = x^2 + 5x$$

$$\frac{dy}{dt} = y^2 + 3y - 1$$

② $x' = y^2 - x^2$ $y^2 - x^2 = 0$
 $y' = x - 1$ $x - 1 = 0 \Rightarrow \underline{x = 1} \Rightarrow y^2 = 1 \Rightarrow \underline{y = \pm 1}$
 $\Rightarrow \underline{[1, 1]}, \underline{[1, -1]}$

$$J = \begin{pmatrix} -2x & 2y \\ 1 & 0 \end{pmatrix}$$

[1, 1]: $J = \begin{pmatrix} -2 & 2 \\ 1 & 0 \end{pmatrix} \Rightarrow \begin{vmatrix} -2-\lambda & 2 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 + 2\lambda - 2 = 0$

$$\lambda_{1,2} = \frac{-2 \pm \sqrt{4+8}}{2} = \frac{-2 \pm 2\sqrt{3}}{2} = \underline{\underline{-1 \pm \sqrt{3}}} \geq 0$$

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[1, -1]: $J = \begin{pmatrix} -2 & -2 \\ 1 & 0 \end{pmatrix} \Rightarrow \begin{vmatrix} -2-\lambda & -2 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 + 2\lambda + 2 = 0$

$$\lambda_{1,2} = \frac{-2 \pm \sqrt{4-8}}{2} = \frac{-2 \pm 2i}{2} = \underline{\underline{-1 \pm i}} \Rightarrow \text{STABILNÍ
OHNISKO}$$